

Cable-Stayed Bridge over River Chenab

Tender Reference: TND-2026-00003 | Issued by: Punjab Roads & Infrastructure Authority (PRIA)

04 — Technical Specifications

Pylon Concrete

Grade C-50 (High Performance Concrete). Low w/c ratio ≤ 0.38 . Silica fume 8–10% by weight of cement. All pylon concrete to be pumped. Maximum chloride content 0.15% by mass of cement.

Pile Concrete

Grade C-40 (Underwater pour — Tremie method). Anti-washout admixture mandatory. Slump ≥ 200 mm. Continuous pour; no cold joints permitted.

Deck Slab Concrete

Grade C-35. Superplasticizer per ASTM C494 Type F.

Reinforcing Steel

ASTM A615 Grade 60 for all RCC. High-yield deformed bars. Epoxy-coated rebars in splash zone (2 m above and below water level).

Stay Cables

Parallel Wire Strand (PWS) system per EN 10138 / PTI DC45.1. Minimum UTS 1,860 MPa per wire. Factory pre-stressing and corrosion protection (grease-filled HDPE pipe). Dampers required on cables $L > 50$ m.

Structural Steel (Main Girders)

ASTM A572 Grade 50 or equivalent. Minimum yield strength 345 MPa. Ultrasonic testing of all welds. Blasting to Sa 2.5 and epoxy-polyurethane paint system (250 microns DFT minimum).

Bearings

Pot bearings per BS EN 1337-5. All bearing pads to be factory-tested. Load capacity and movement capacity to match design loads.

Deck Waterproofing

Bonded waterproofing membrane (spray-applied polyurea, 2.5 mm min.) to full deck slab prior to wearing course.

Wearing Course

Modified bitumen asphalt (SBS-modified), 50 mm compacted thickness. Stone mastic asphalt (SMA) optional subject to Engineer approval.

Scour Protection

Quarry stone pitching minimum 600 mm thick, $D_{50} \geq 300$ mm. Geotextile filter layer underneath. Anti-scour mattresses minimum 300 kg/m².